

Batch Quality Risk Report

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Summary

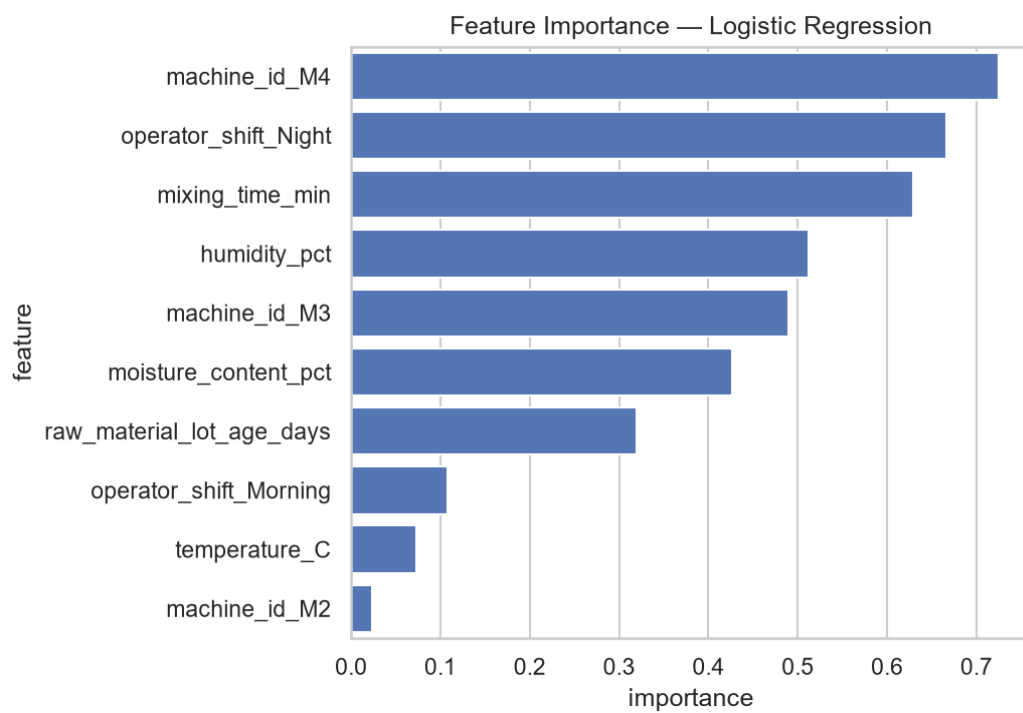
Of **20** incoming batches scored, **9** were flagged for manual QC review before proceeding further in the process (9 High risk, 10 Medium risk, 1 Low risk). Flagging is based on a model-estimated failure probability, using a decision threshold tuned to prioritize catching true failures over minimizing false alarms.

Batches Flagged for Review

Batch ID	Fail Prob (%)	Risk	Machine	Shift	Mixing (min)	Humidity (%)
NEW018	95.6	High	M1	Morning	60.8	68.6
NEW016	85.7	High	M1	Night	52.6	60.2
NEW020	84.4	High	M1	Afternoon	50.5	56.1
NEW002	82.9	High	M2	Morning	50.0	52.0
NEW005	79.8	High	M1	Morning	50.2	60.3
NEW008	74.6	High	M3	Morning	49.8	56.5
NEW017	67.5	High	M4	Night	46.5	40.7
NEW001	64.3	High	M3	Afternoon	66.7	40.1
NEW010	55.9	High	M2	Night	50.9	43.9

What Drives These Predictions

The model weighs several process factors together. Historically (in the training data), the strongest signals were: machine M4, night shift operation, longer mixing times, and higher humidity. The chart below shows relative feature importance from the trained model.



Appendix: Model Performance

For reference, this report is generated by a Logistic Regression model (selected over a Random Forest alternative for better ROC-AUC), evaluated on a held-out test set of real (synthetic) outcomes:

Metric	Value
ROC-AUC	0.706
Accuracy	0.720
Precision (Fail)	0.374
Recall (Fail)	0.627
Decision threshold	0.553

Note: built on synthetic data designed to reflect realistic manufacturing relationships, as a demonstration of the full pipeline -- not on real plant data.